

The Shifting Wall & Slipping Base Challenge

Role: Robotics Research Resident – Localization & Mapping (SLAM)

1. Overview & Context

This assignment simulates a high-entropy, degenerate residential construction site environment. As a candidate, you will build a simulation stress-test environment, inject non-idealities, and implement core state estimation, localization metrics, and lifelong mapping filters from first principles. You are expected to complete this assignment using Gazebo or Isaac Sim without specialized GPU hardware.

2. Assignment Tasks

Task 1: Simulation Setup & Degenerate Corridor

- **Environment:** Construct a programmatically generated 20-meter straight corridor with flat, featureless concrete walls to introduce structural axial degeneracy.
- **Shifting Wall:** Place a large wall panel halfway down the hall that shifts sideways by 1.5 meters precisely at $T=15$ seconds intervals during execution.
- **Robot Model:** Configure a differential drive robot with a 16-channel 3D LiDAR and a 6-axis IMU.

Task 2: In-Situ Noise Injection & Sensor Fusion

- **Noise Node:** Write a standalone node that intercepts ground-truth odometry and applies the mathematical models detailed in Section 3 of this document to simulate dynamic floor slip and sensor deterioration.
- **Fusion & Analytics:** Implement or rigorously tune an EKF/Factor Graph pipeline. Output the running pose covariance matrix and calculate the Information Matrix of scan matching to track axis degeneracy.

Task 3: Lifelong Map Updates & Diagnostics

- **Space Clearing:** Implement ray-casting or sub-mapping logic to clean the 'ghost obstacle' artifact left behind when the dynamic panel moves.
- **Diagnostics:** Detects when LiDAR alignment uncertainty along the corridor axis spikes, logging a clear `LOCALIZATION_DEGENERACY_WARNING`.

3. Mathematical Framework for Noise Injection

Candidates must implement the following mathematical models within their noise injection node to simulate structural non-idealities:

A. Linear Odometry Slip Model (Dynamic Intermittent Scale Slip)

To simulate loose plaster dust and physical tire slip on construction floors, the recorded forward linear velocity (v_{recorded}) must scale downward while the actual physical simulator joints spin at full speed. Implement a state-dependent multiplicative noise model:

$$v_{\text{recorded}}(t) = s(t) * v_{\text{true}}(t) + w_v(t)$$

Where:

- $v_{\text{true}}(t)$ is the actual ground truth velocity from the physics engine.
- $w_v(t) \sim N(0, \sigma_v^2)$ is typical high-frequency white Gaussian measurement noise.
- $s(t)$ is a dynamic scaling factor governed by an explicit Bernoulli trial or step function simulating patch-based slipping:
- $s(t) = 0.70$ if $10 \leq \text{Position}_X \leq 15$ (30% slip over a 5m dusty floor patch) $s(t) = 1.00$ otherwise

B. IMU Sensor Deterioration Model (Random Walk + Bias Drift)

To evaluate first-principles filtering robustness, candidates must inject slow-drifting bias on the gyroscope data rather than pure white noise, simulating high-vibration environmental wear:

$$\omega_{\text{measured}}(t) = \omega_{\text{true}}(t) + b(t) + \eta_g(t) \quad b_{\text{dot}}(t) = \eta_b(t)$$

Where:

- $\eta_g(t) \sim N(0, \sigma_g^2)$ represents Gyroscope White Noise (Angle Random Walk).
- $b(t)$ represents the running slow-drifting operational bias.
- $\eta_b(t) \sim N(0, \sigma_b^2)$ represents Bias Instability (Rate Random Walk).

4. Deliverables

- Source Code Repository containing the simulation world, robot URDF, noise node, and estimation pipeline.
- A recorded video snippet showing the execution run and the response to the moving wall panel.
- A concise summary plot tracking: Ground Truth X-Y Trajectory vs. Slipping Unfused Odometry vs. Your Fused State Estimate.
- Output log containing the timestamped state covariance matrices and triggered degeneracy warnings.